



NO GREEN-TICK THEATRE

Incident survival assessment 01-08

Status is bounded to tested synthetic behavior. Each card names the code evidence, first user-visible failure, implemented control, and what still needs external authority.

01

Phone-only patient access

SURVIVES - LOCAL

WHERE backend/app/access.py:61**FIRST BREAK**

A real patient still fails if the production channel broker, recovery policy, or device secret is absent.

BUILT + RESIDUAL BOUNDARY

One-use, hashed, rate-limited channel claim; device-bound patient session; server re-fetches a patient-only projection. Email is never identity. Production assurance level remains an owner decision.

02

One-line tenant isolation defect

PARTIAL

WHERE deployment/postgres/tenant_rls.sql:35**FIRST BREAK**

SQLite would rely on application predicates; one missed predicate can disclose rows reachable by that route.

BUILT + RESIDUAL BOUNDARY

Central actor/object policy, cross-tenant concealment tests, RLS for every tenant table, and FORCE RLS target. It survives only after PostgreSQL migration and non-owner runtime-role verification.

03

PHI through logs and telemetry

SURVIVES - SCOPED

WHERE backend/app/telemetry.py:24**FIRST BREAK**

An unregistered SDK, host proxy, crash agent, or deployment access log can create a new egress path.

BUILT + RESIDUAL BOUNDARY

Low-cardinality allow-list telemetry, route templates, normalized request IDs, metadata-only audit and synthetic identifier canary tests. Third-party dashboards and retention require deployment inspection.

04

Redaction before every model call

SURVIVES - LOCAL

WHERE backend/app/scribe.py:233**FIRST BREAK**

A future provider call outside the typed gateway would bypass the proof.

BUILT + RESIDUAL BOUNDARY

Identifiers are removed and clinical anchors compared before RedactedPayload exists; ProviderGateway accepts only that type. Provider spies prove zero calls on fidelity failure. Network egress policy is not deployed.

05

Clinic B onboarding

SURVIVES - DEMO

WHERE backend/app/configuration.py:41**FIRST BREAK**

Production onboarding fails on missing secrets, regional contracts, migration state, or policy ownership.

BUILT + RESIDUAL BOUNDARY

Versioned locale, channel, provider, timeout, retention, feature and escalation configuration; second clinic works without code edits; rollback and cross-clinic cache/policy tests. Secret provisioning stays external.

06

Trilingual noisy consult

PARTIAL

WHERE backend/app/capture.py:285**FIRST BREAK**

The browser records audio but no live ASR adapter turns it into transcript events; real Hokkien can be mistranscribed or omitted.

BUILT + RESIDUAL BOUNDARY

Provider-neutral timestamped language spans, speaker label, ASR/audio quality and explicit unsupported-language abstention. No WER/CER or safety-concept recall claim without licensed SEA audio and native review.

07

Allergy at minute two

SURVIVES - SIMULATED

WHERE backend/app/capture.py:219**FIRST BREAK**

Real detection latency and sensitivity fail with an inaccurate or delayed ASR adapter.

BUILT + RESIDUAL BOUNDARY

Idempotent ordered segment stream emits a provisional, timestamp-bound allergy signal before finalization; unsupported spans abstain; clinician confirm/dismiss is audited. The test is event-level, not real audio.

08

Provider hangs for 45 seconds

SURVIVES - SIMULATED

WHERE backend/app/providers.py:137**FIRST BREAK**

A provider that ignores cancellation can consume worker capacity after the user has received a fallback.

BUILT + RESIDUAL BOUNDARY

Server deadline, non-blocking cancellation, circuit breaker, stable failure code and deterministic rule-only draft; UI labels degradation. Production needs process isolation, concurrency budgets and queue monitoring.



NO GREEN-TICK THEATRE

Incident survival assessment 09-16

Status is bounded to tested synthetic behavior. Each card names the code evidence, first user-visible failure, implemented control, and what still needs external authority.

09

Provider returns 503 for one hour

SURVIVES - SIMULATED

WHERE backend/app/providers.py:146**FIRST BREAK**

Without a durable production job queue, automatic recovery work can be lost across process restart.

BUILT + RESIDUAL BOUNDARY

Repeated failures open the circuit; deterministic evidence remains and new drafts are visibly rule-only or fail closed. No stale AI result is relabelled as current. Durable retry infrastructure is still external.

10

Concurrent note edits

SURVIVES - CONTRACT

WHERE backend/app/care.py:196**FIRST BREAK**

True character-level live coauthoring is not supported; disconnected local drafts still depend on browser persistence.

BUILT + RESIDUAL BOUNDARY

Expected-version write prevents lost update; 409 returns base/current/proposed snapshots; deterministic three-way merge is only a reviewed draft and cannot save until explicitly compared.

11

Appointment link never arrives

SURVIVES - SIMULATED

WHERE backend/app/delivery.py:450**FIRST BREAK**

The demo admin records provider receipts; no contracted channel or signed webhook proves live delivery semantics.

BUILT + RESIDUAL BOUNDARY

Purpose-specific outbox distinguishes queued, accepted, delivered, patient acknowledged, failed and overdue. Failed/overdue creates one owned care task; late acknowledgement closes it without erasing history.

12

Wrong dosage reaches patient

PARTIAL

WHERE backend/app/terminology.py:104**FIRST BREAK**

A clinician can still approve a clinically wrong but syntactically supported dose; the local vocabulary is not a drug-safety authority.

BUILT + RESIDUAL BOUNDARY

Exact source offsets, unit normalization, unresolved-dose block, medication-specific attestation, immutable approval receipt and correction delivery. RxNorm/local formulary and clinical validation remain production gates.

13

Conflicting allergy assertions

PARTIAL

WHERE backend/app/conflicts.py:87**FIRST BREAK**

Unseen phrasing, temporality, certainty or language may evade the bounded deterministic detector.

BUILT + RESIDUAL BOUNDARY

Safety-floor conflict card presents two immutable sources side by side; reviewer must inspect both, record rationale, or escalate unresolved. No latest-note or source-hierarchy auto-winner.

14

Meaningless confidence number

SURVIVES - DEFINED

WHERE backend/app/importance.py:35**FIRST BREAK**

A user can still misread evidence support as probability if explanatory copy is removed later.

BUILT + RESIDUAL BOUNDARY

Evidence support is a versioned trust-state mapping, never model self-confidence; bands, decomposition, error tests and fail-safe behavior are visible. Unsupported review questions abstain.

15

Exposure bias and fatigue

PARTIAL

WHERE backend/app/evaluation.py:37**FIRST BREAK**

Deterministic surfaced-only data cannot estimate unseen outcomes or justify promotion.

BUILT + RESIDUAL BOUNDARY

Feedback changes a shadow score only; server logs propensity; hard allergy/medication/urgent floors resist dismissal; evaluation refuses promotion without overlap, ESS and uncertainty evidence. No live causal claim.

16

Edited source behind a citation

SURVIVES - LOCAL

WHERE backend/app/provenance.py:60**FIRST BREAK**

A deleted/corrupted store or untracked external source cannot be reconstructed by the application hash alone.

BUILT + RESIDUAL BOUNDARY

Citation remains bound to immutable version, offsets, quote and hash; current edit marks dependency stale and shows original/current side by side. Regeneration creates a proposal and preserves protected human state.



RESEARCH AND INNOVATION BOUNDARY

Capability group 17: broad contracts, narrow claims

The build integrates the requested stack end to end, but deliberately marks real audio, clinical validation, provider transport, legal interpretation, and deployment controls as external evidence gates.

Selected system design: evidence before automation

1Identity

2Stream

3Privacy

4Evidence

5Human gate

6Delivery

STREAMING + NOISY ASR

Ordered resumable segment contract and quality abstention; no live ASR/WER corpus.

PARTIAL

SPEAKER ATTRIBUTION

Provider label is evidence, never biometric identity or clinical authority; human correction retained.

PARTIAL

CODE SWITCHING

Within-statement language spans; unsupported Hokkien abstains instead of becoming generic Chinese.

PARTIAL

MULTILINGUAL DOWNSTREAM

Original spans remain primary; safety extraction handles bounded English terms only.

PARTIAL

TERMINOLOGY + DOSAGE

Exact local evidence and fail-closed units; RxNorm/formulary validation is not claimed.

PARTIAL

IMMUTABLE PROVENANCE

Version, offsets, quote, URI, hash, stale propagation and original/current comparison.

SURVIVES

NEGATION + CONFLICT

Bounded deterministic candidates and two-source review; unknown semantics stay unresolved.

PARTIAL

COLLABORATIVE EDITING

Optimistic concurrency plus reviewed three-way draft; no CRDT claim.

SURVIVES

AI REGENERATION

New proposal only; human confirmations, tasks, conflict decisions and deliveries preserved.

SURVIVES

AUDIENCE OUTPUTS

Server allow-list projections; AI drafts internal; patient release requires clinician confirmation.

SURVIVES

BOUNDED SELF-LEARNING

Clinic/role shadow posteriors, hard floors and promotion refusal under exposure bias.

PARTIAL

PATIENT DELIVERY

Immutable approved copy, receipt state, patient acknowledgement, correction and escalation.

SURVIVES

WHAT I TRIED

Tested deadline, 503, low-quality stream, unsupported language, stale citation, concurrent edit, contradiction, wrong dose, failed delivery and late acknowledgement paths - including desktop and mobile browsers.

WHERE I STOPPED

No licensed SEA clinical audio, native-speaker labels, approved drug reference, real messaging provider, production OIDC/RLS runtime, legal ruling, clinical reviewer, or prospective pilot evidence was available. The UI names these absences.

ASSUMPTIONS THAT REMAIN

All identities and care facts are synthetic. Provider and delivery callbacks are deterministic rehearsals. Hashes prove application consistency, not external notarization. A passing test is not clinical safety or compliance certification.

100% / 100%
BACKEND / FRONTEND COVERAGE

13 / 13
DESKTOP + MOBILE BROWSER

2,502 / 2,502
KILLED / CHECKED MUTANTS

2.188 ms
LOCAL WARM-PATH P95

0
KNOWN ADVISORIES AT SCAN

SELECTED PRIMARY REFERENCES - HYPERLINKED

[1] NIST SP 800-63B identity	[2] PostgreSQL row security	[3] OWASP logging guidance
[4] OpenTelemetry sensitive data	[5] HL7 FHIR Provenance	[6] HL7 FHIR Communication
[7] HL7 FHIR Appointment	[8] NLM RxNorm APIs	[9] FDA CDS guidance, Jan 2026
[10] Singapore PDPC healthcare	[11] AWS transactional outbox	[12] W3C WCAG 2.2

Synthetic data only. Prototype - not for clinical use or a compliance claim.

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